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Research Insights

Double polymerization synergy to sharpen structural toughness of melamine resin foam

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ABSTRACT

Melamine resin foam (Mfoam) is a porous heat-cured amino resin material with excellent flame retardancy, thermal insulation, and noise reduction capabilities. However, some fatal shortcomings like easy brittle breakage, easy powderization and poor toughness directly limit its wide application and further development. In this paper, hydroxypropyl acrylate (HPA) was used to participate in the synthesis of melamine resin oligomers (Moli) in order to link the vinyl group to molecular skeleton. As a result, both vinyl and hydroxymethyl active groups are present in Moli which provide Moli with the ability to perform free radical polymerization reaction and polycondensation polymerization reaction, named double polymerization reaction (DPR), and produce synergistic effects to form a structurally toughened melamine resin foam (TMfoam). The research shows that the scheme can effectively solve the problems of brittle breakage and easy powderization of Mfoam. And under apparent density (Ad) of 40 ± 2 kg/m^3, the prepared TMfoam has good toughness, with the bending strength of 405.2 kPa, the compression strength of 352.5 kPa, and the pulverization rate of 1.18 %. Meanwhile, the combustion grade of the TMfoam can reach to B_1 grade, and the thermal conductivity is 0.0328 W/(m·K). Also the absorption coefficient of TMfoam can reach up to 0.92 under frequency of 3045 Hz. All the facts indicate that the DPR showed good synergistic effect during the forming and setting stages from Moli to TMfoam. At the same time, the bubble chamber of TMfoam changes from metastable state to stable state, and its strength and toughness are significantly improved in step. This research has great and practical significance for expanding the wide application of Mfoam type materials.

1. Introduction

Melamine resin foam (Mfoam) is a thermosetting porous material, that is prepared by foaming process of melamine resin oligomers (Moli) under certain conditions [1–3]. During Mfoam forming process, the molecular weight of Moli increases rapidly through polymerization reaction of hydroxymethyl groups, or say cross-linking reaction, to form a three-dimensional network polymer foam [4–6]. Generally, the Mfoam presents excellent intrinsic flame retardant properties [7–9], that is an unique feature different from all the other polymer foams [10], such as polyurethane foams [11,12], urea–formaldehyde foams [13], phenolic foams [14], polystyrene foams [15], etc. This feature of Mfoam attracts great interest and has become a hot spot in research of porous foam materials.

On the other hand, the Mfoam has obvious defects such as easy brittle breakage, easy powderization and poor toughness, which seriously restricts further technical development [16,17]. To solve these problems, a large number of researches have been carried out. For example, Li [18] et al. prepared a rigid closed-cell foam with melamine formaldehyde (MF) using ethylene glycol (EG) and the carbon fiber (CF) as a composite

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toughening agent with the apparent density 63.22 kg/m^3, the pulverization rate of 5.21 %, the compressive strength of 355.3 kPa, bending strength of 0.44 MPa, closed-cell ratio of 79.1 %, water absorption of 9 %, thermal conductivity of 0.031 W/(m·K), and limiting oxygen index (LOI) of 39.6 %. Li [19] et al. modified melamine resin with benzoguanamine (BG), when the BG content was 15 %, its energy storage modulus increased by 18.6 % compared with unmodified melamine resin, and its LOI increased from 36.2 % to 38.4 %, indicating that the addition of BG enhanced its flame retardant performance. Yan [20] et al. modified the melamine resin with allyl glycyl ether to prepare a foam material with an apparent density of 250 kg/m^3, the compressive strength of 5.08 MPa, the limiting oxygen index of 38.9 %, thermal conductivity of 0.0256 W/(m·K), and the volumetric water absorption of 0.9 %.

In all above studies, the main idea is to optimize the structure of Moli by a physical or chemical modification to enhance the performance of Mfoam. This does improve the performance of Mfoam materials in some aspects, but it still does not fundamentally solve the defects such as brittle fracture, easy pulverization and poor toughness.

The molecular structure of melamine shows an obvious rigid nature. The Mfoam derived from Moli is a multicell polymer of high crosslinking density formed by polymerization reaction, which leads to the inherent defects of easy cracking, fragility and low strength of Mfoam, this also is its basic properties.

In this paper, hydroxypropyl acrylate (HPA) was incorporated into the synthesis process of Moli so as to link the vinyl group to its molecular skeleton. Thereafter, there are both vinyl and hydroxymethyl active groups in Moli, and it provide Moli with the ability to perform free radical polymerization reaction and polycondensation polymerization reaction, named double polymerization reaction (DPR). Further, the synergistic effect of DPR was promoted by reasonable regulation to Moli foaming process, to optimize the molecular structure of Mfoam, and prepare a new TMfoam with excellent structural toughness and strength. The synergistic effect of DPR is beneficial to effectively solve the shortcomings of Mfoam, and it is a valuable technical solution for developing new high-performance intrinsic flame-retardant functional Mfoam.

2. Experimental part

2.1. Raw material

Melamine (M, 99 %), polyformaldehyde (95 %), hydroxypropyl acrylate (HPA, 97 %), diethanolamine (99 %), Tween-80 (T-80), sodium persulfate (99 %), n-hexane (97 %), etc. (analytically pure), purchased from Macklin reagent.

2.2. Preparation of Moli and TMfoam

According to the stoichiometric ratio, M, paraformaldehyde, water, and diethanolamine were added to a 1 L three-necked flask equipped with a stirrer, thermometer, and condenser. The temperature was raised to 100 °C to complete the hydroxymethylation reaction (HR). After M was completely dissolved and the water solubility tolerance point appeared, the temperature was lowered to 70 °C, and HPA was added as the etherification reagent to complete the etherification reaction (ER). The reaction continued for 1.5 h, The Moli with vinyl and hydroxymethyl active groups were obtained.

Both vinyl and hydroxymethyl active groups exist in the molecular structure of the Moli. During the foaming process, the two reactive groups underwent deep polymerization, significantly increasing the polymerization degree and molecular weight of the melamine resin, and completing the forming and setting of the TMfoam.

In the experiment, the general foaming scheme [21–24] was adopted. The foaming solution was prepared by uniformly mixing Moli (200 g), T-80 as emulsifier (1.5–3.5 g), n-hexane as blowing agent (1.5–3.5 g), and sodium persulfate as initiator (1–3 g). The TMfoam were prepared by the two stages: the foam forming at 30–80 °C and foam setting at 100–150 °C, as shown in Fig. 1.

2.3. Characterization and testing

(1) The infrared (IR) spectrums were recorded on a Fourier Transform Infrared Spectrometer (Shimadzu IRAffinity-1S, Japan). About 5 g of each sample was dried at 105 °C for 5 h in an oven. The samples (Moli) were ground to a fine powder and then measured under 32 scans in the range 400–4000 cm^{-1} with a resolution of 4 cm^{-1}.
(2) The viscosity of Moli were determined by digital portable viscometer (TY-NDJ5S, China).
(3) Thermogravimetric (TG) analysis and derivative thermogravimetry (DTG) were measured on a thermogravimetric (Discovery TGA 550, America). Under nitrogen atmosphere, about 2–3 mg of each sample was at the temperature range 30–800 °C and the heating rate was 10 °C/min.
(4) The compressive and bending strength of the TMfoam were tested by an electronic universal material testing machine (CMT4204, China).
(5) The properties of the TMfoam were tested with reference to the following criteria:ISO845:2006 "Cellular plastics and rubbers-Determination of apparent density", GB/T8813-2008 "Test method for compression of rigid cellular plastics", GB/T12812-2006 "Rigid cellular plastics—Determination of friability", GB/T 5486-2008"Test methods for inorganic rigid thermal insulation", GB/T 2406-1993"Plastics-Determination of flammability by oxygen index", ISO 22007-2-2015"Plastics-Determination of thermal conductivity and thermal diffusivity – Part 2:Transient plane heat source (hot disc) method", GB/T2408-2008"Plastics-Determination of burning characteristics-Horizontal and vertical test", GB/T 8624-2012"Classification for burning behavior of building materials and products".

3. Results and discussion

3.1. The preparation and structural characteristics of Moli

3.1.1. The optimization of the preparation conditions for Moli

Extensive experiments have confirmed that when the solid content of Moli is less than 70 %, it is difficult to apply it to foam forming. However, when the solid content is more than 75 %, it is suitable for foam forming, and the appearance of the TMfoam is intact.

The dosage of HPA (mole ratio) used in the preparation of Moli was regulated, and the basic physical properties of different Moli were compared, as shown in Table 1.

As shown in Table 1, with the increase in HPA content, the solid content and viscosity of Moli gradually decreased, while the stability time gradually increased. The results showed that a sufficient dosage of HPA was beneficial to reduce the viscosity of Moli. The reasonable dosage of HPA contributed to inhibiting the crosslinking rate and crosslinking density of hydroxymethyl in Moli.

The preparation of Moli should not only consider appropriate solid content, viscosity, and stability, but also take into account the adjustability of the foaming process [23,24]. When the viscosity of the Moli is too high (HPA mole ratio $\leq$ 0.3 mol), it is difficult to achieve uniform mixing during the preparation of the foaming solution, and cracks appeared in the TMfoam. While the viscosity of Moli is too low (HPA mole ratio $\geq$ 0.6 mol), it is prone to shrinkage and collapse during the preparation of TMfoam.

It is evident that appropriate solid content, viscosity, and stability time are essential for achieving good foamability in Moli. As demonstrated in Table 1, when the HPA mole ratio is within the range of 0.4–0.5 mol, the solid content >75 %, the viscosity range of 810–965

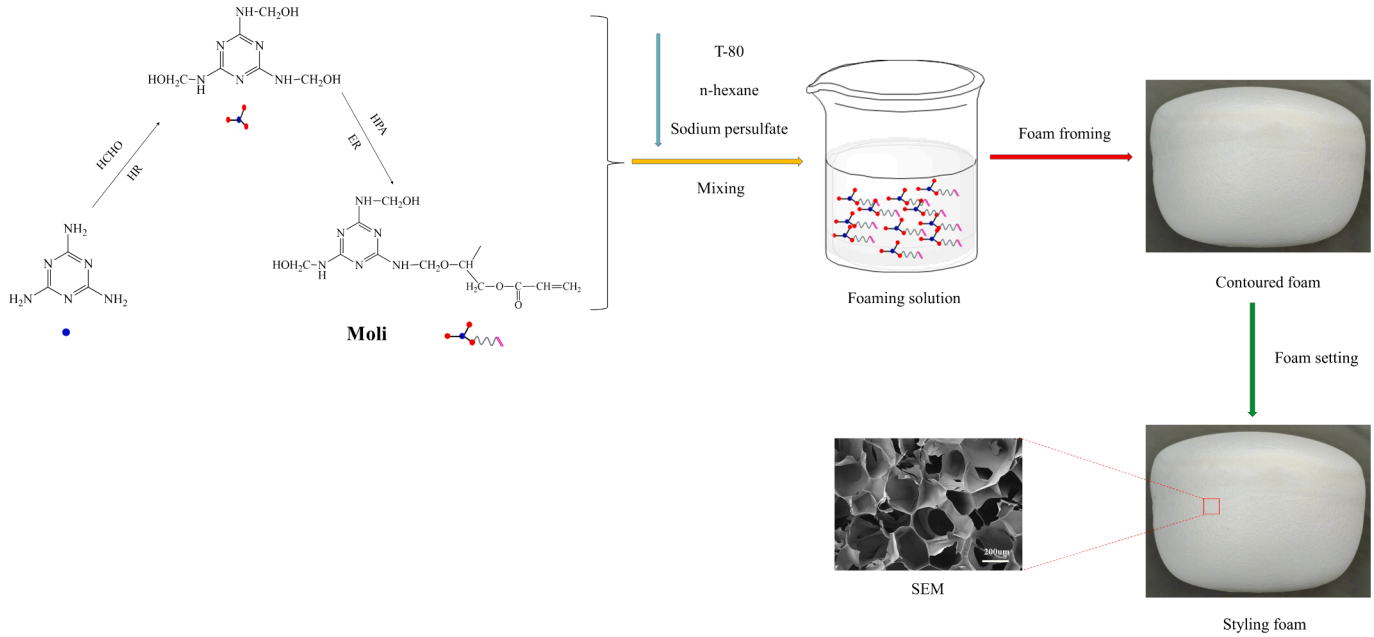

Fig. 1. Schematic diagram of preparation of Moli and TMfoam.

Table 1
The basic physical properties of different Moli.

Sample number	Material ratio	Appearance	Solid content (%)	Viscosity (mPa·s)	Stabilization time (h)	Foam appearance
#1	1:3.0:0	Transparent and sticky liquid.	82.4	1460	2	Crack
#2	1:3.0:0.1		80.6	1320	5	Crack
#3	1:3.0:0.2		78.9	1235	12	Crack
#4	1:3.0:0.3		76.3	1036	20	Ok
#5	1:3.0:0.4		75.6	965	48	Ok
#6	1:3.0:0.5		72.9	816	56	Ok
#7	1:3.0:0.6		69.1	622	63	Collapse
#8	1:3.0:0.7		67.6	494	72	Collapse
#9	1:3.0:0.8		65.2	425	96	Collapse

Note: Material ratio refers to n (M):n (polyformaldehyde):n (HPA) = 1:3.0: (0–0.8).

mPa·s, and the stability time ≥ 48 h, the Moli exhibits favorable foamability.

3.1.2. Infrared structural characteristics of Moli

The structural characteristics of the Moli without HPA participating in the etherification reaction (sample #1) and the Moli used for foaming (sample #5 and sample #9) were compared, as shown in Fig. 2 [25,26].

In Fig. 2, as the dosage of HPA increases, there are four characteristic absorption peaks in the structure of Moli, and the intensity of the absorption peaks gradually increases. The absorption peak at 2865 cm^{-1} is attributed to methyl stretching vibration, 1720 cm^{-1} belongs to the carbonyl stretching vibration peak, the carbon–carbon double bond telescopic vibration absorption peak at 1650 cm^{-1}, and the C—O—C absorption peak at 1050 cm^{-1}. This indicates that HPA was involved in the reaction process of the Moli, or that there are vinyl and hydroxymethyl groups in the molecular structure of the Moli used for foaming.

The other characteristic peaks of the Moli are classified as follows: —OH and —NH stretching vibration peaks are located at 3400 cm^{-1}, —CH_2— stretching vibration peaks are located at 2950 cm^{-1}; 1560 cm^{-1} belongs to the vibrational absorption peaks of secondary amine group in the triazine ring (—N—C═N); 1162 and 1003 cm^{-1} are the stretching vibration peaks of methylene ether bond (—CH_2—O—); and 813 cm^{-1} is the characteristic stretching vibration absorption peak of triazine ring.

It can be seen that HPA participates in the reaction of the Moli and introduces the vinyl group into its molecular structure. This is the structural basis for the DPR to form TMfoam, which is of great significance for improving the properties.

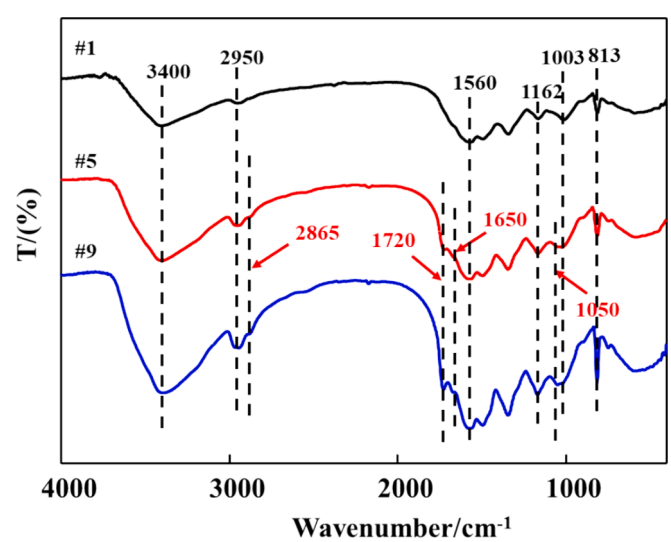

Fig. 2. IR spectra of different Moli samples.

3.1.3. Thermal characteristics of Moli

The thermal stability of the above three samples [27] was compared

and analyzed, as shown in Fig. 3 and Table 2 respectively.

From Fig. 3 and Table 2, there are slight differences in the thermal stability of the samples prepared with different dosages of HPA, but the overall change is not obvious. With the dosage of HPA increasing, the corresponding maximum weight loss temperature of the samples slightly decrease, that is, from 359.2 °C (0 mol HPA) to 357.2 °C (0.4 mol HPA) and then to 356.0 °C (0.8 mol HPA). This indicates that the change in HPA mole ratio has little effect on the thermal stability of the Moli. It can be seen that the dosage of HPA has little influence on the foaming process and the thermal stability of foam forming.

3.2. Regulation of foaming process of DPR

There are many controllable factors [2] that resulting in the foam forming of Moli, including the dosage of blowing agent and initiator, foam forming temperature, foam setting temperature and time, etc. In order to effectively reduce the complexity of foam forming regulation, simplify the foaming process, and facilitate the analysis of key regulatory factors.

Take sample #5 as an example, the general foaming scheme was slightly adjusted for applicability, and the main control parameters affecting the Ad of the TMfoam were compared and discussed.

3.2.1. The temperature effect on the DPR and foaming process

The Moli contain two groups of vinyl and hydroxymethyl group, and the DPR have significant differences: the reaction between vinyl group is free radical polymerization reaction [28,29], while the reaction between hydroxymethyl groups is polycondensation reaction, which is an important feature of the foaming process. In the foaming process of DPR, linear structure segments and cross-linked structures are formed respectively, and the linear segments are laid or interspersed in the cross network grid, ultimately improving the structural toughness of the TMfoam.

The temperature conditions have a direct impact on the foaming process. The foaming process of Moli for foaming includes two temperature stages: foam forming temperature and foam setting temperature.

Taking the regulation of the two temperature segments during the preparation of TMfoam (sample #5) as an example, the influence of temperature conditions on the Ad of TMfoam was shown in Fig. 4.

As shown in Fig. 4a and 4b, when the dosages of initiator and blowing agent are the same, the foam forming time and foam setting time gradually decreases with the increase of the temperature of the two temperature segments, and the Ad of the TMfoam shows a trend of first decreasing and then increasing. In a reasonable temperature range, the Ad of TMfoam can reach a lower extreme value. At low temperature, the DPR proceed slowly, and the heat generated by the system is not conducive to the expansion of bubble, resulting in a higher Ad of the TMfoam. When the temperature is high, the DPR accelerate, and the heat generated by the system causes the bubble to expand rapidly, or even burst. At the same time, during the process of bubble expansion, it may be difficult for the bubble wall to form sufficient support in terms of thickness and strength, resulting in the collapse of the bubble wall due to local damage, which is not conducive to reducing the Ad of the TMfoam. It can be seen that only under reasonable temperature conditions can the volume of the bubble be maximized, and the strength of the bubble wall can form a good support, thereby ensuring the overall stability of the TMfoam and the minimum Ad. The experimental results show that the foam forming temperature and foam setting temperature is 50 °C and 130 °C respectively, and the Ad of the TMfoam about 40 kg/m^3, which can better meet the requirements of practical application.

Based on this, it is beneficial to further optimize the foaming conditions to prepare TMfoam with excellent overall appearance and comprehensive properties.

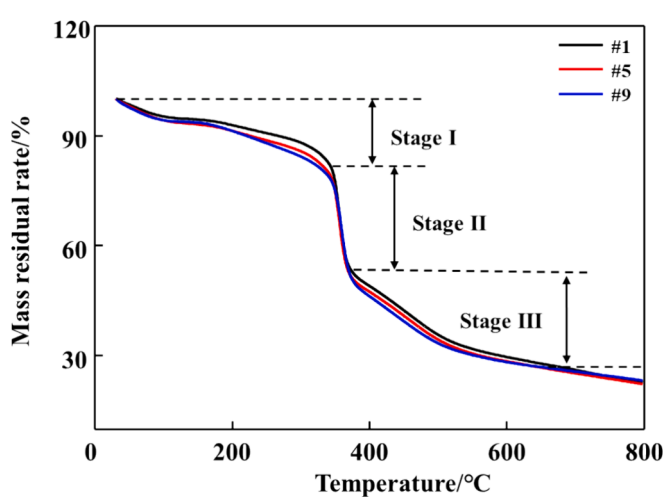

Fig. 3. Thermogravimetric curves of Moli.

3.2.2. The dosages of initiator and blowing agent on the DPR and foaming process

The foaming of Moli is a synergistic process of DPR. The dosages of initiator and blowing agent are the two key factors affecting the Ad of TMfoam. Therefore, the dosages of initiator and blowing agent have a significant impact on the foaming process of the Moli.

Taking the preparation process control of TMfoam (sample #5) as an example, the influence of the dosages of initiator and blowing agent on the Ad of TMfoam was analyzed respectively at the foaming forming temperature and foam setting temperature of 50 °C and 130 °C, as shown in Fig. 5.

In Fig. 5, with the dosages of initiator and blowing agent increasing, there is a minimum Ad of TMfoam. This indicates that a reasonable dosage of initiator and blowing agent is beneficial to the successful implementation of the foaming process of the Moli. When the dosages of initiator and blowing agent are 1.00 % and 1.25 % respectively, the Ad of the TMfoam is approximately 40 kg/m^3, and a lower Ad is conducive to the lightweight of the foam material.

From the above, it can be seen that when the composition and proportion of the foaming solution is reasonable, the dosages of the initiator and blowing agent are appropriate, the requirements for the Ad of the foam under practical application can be well met. On this basis, through the rational control of the foaming conditions, it is beneficial to realize the optimization of the comprehensive properties of the TMfoam.

3.2.3. Morphological characteristics of TMfoam

The prepared TMfoam with an Ad of 40 ± 2 kg/m^3 were presented in Fig. 6.

In Fig. 6, foam solutions were prepared by using the Moli with HPA dosages of 0.1 mol and 0.8 mol, resulting in the obtaining of TMfoam sample #2 and sample #9, respectively. Among them, sample #2 has obvious macroscopic cracks and sample #9 has macroscopic collapse, and both samples have more fragments and holes in the microscopic morphology.

Compared to sample #2 and sample #9, the foaming solution prepared by using the Moli synthesized with 0.4 mol HPA, corresponding to sample #5, is uniform and regular inside and outside the foam without defects. At the same time, the size of the micropore is uniform, and there are many closed-cell structures. This also confirms the good effect of foam regulation through the key regulatory factors mentioned above.

In summary, when the dosages of the initiator (1.00 %), blowing agent (1.25 %), the foam forming temperature and foam setting temperature is 50 °C and 130 °C, respectively, the preparation of TMfoam with an apparent density of approximately 40 kg/m^3 can be achieved perfectly.

Table 2
Thermogravimetric data of Moli.

Sample	Thermal weight loss stage Temperature interval (°C)/Weight loss ratio (%)			Mass residual rate (%)	Maximum weight loss temperature (°C)
	I	II	III		
#1	31.5–342.1 (17.3)	342.1–375.3 (29.5)	375.3–800 (29.6)	21.2	359.2
#5	31.5–338.5 (18.9)	338.5–379.1 (29.9)	379.1–800 (29.0)	22.2	357.3
#9	31.5–337.1 (21.0)	337.7–380.3 (30.4)	380.3–800 (27.5)	23.6	356.0

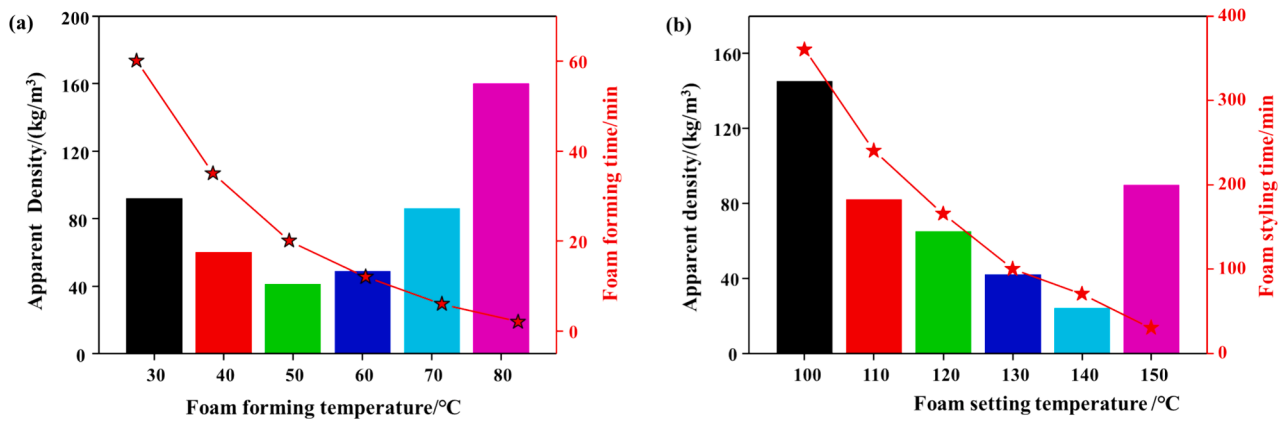

Fig. 4. Relationship between temperature and Ad of TMfoam. Note: Moli(200 g), T-80(2.0 g), Initiator (1.8 g), blowing agent (2.0 g).

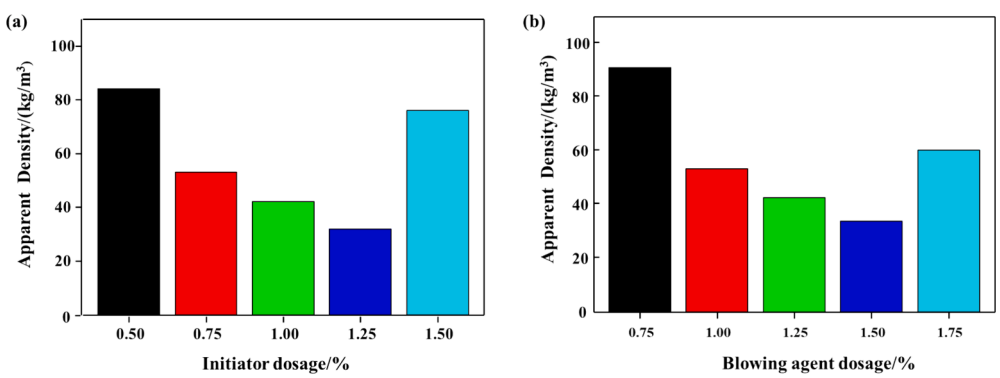

Fig. 5. Relationship between initiator-blowing agent dosages and Ad of TMfoam. Note: Moli(200 g), T-80(2.0 g).

3.3. The impact of DPR on the properties of TMfoam

By optimizing the control parameters for foam forming and foam setting, uniform TMfoam with micro-vesicle-sized and multi-closed-cell structures can be successfully obtained. In practical applications, the mechanical properties of TMfoam, especially flexibility, need to be considered, as it is the basic guarantee and requirement for the practicality of foam material.

3.3.1. The toughness performance of TMfoam

The pulverization rate and the basic mechanical properties (bending strength and 10 % compression strength) of the TMfoam prepared by adjusting the parameters of the foaming process were tested and analyzed, as shown in Fig. 7.

As shown in Fig. 7a, with the dosage of HPA increasing, the pulverization rate of TMfoam generally showed a downward trend. However, after a significant decline to the lowest point, there was a slight increase, which may be due to the fact that in the later stage of foam forming, although the number of linear molecular chains increased, there was also a bubble bursting phenomenon.

In Fig. 7b, both the bending strength and 10 % compression strength of TMfoam increased first and then decreased with the increase of HPA content, and had a positive correlation with the Ad value. Taking the foam with HPA content of 0.4 mol as an example (sample #5). When the Ad was 40 ± 2 kg/m^3, the bending strength could reach to 405.2 kPa and the 10 % compression strength was 352.5 kPa; while when the Ad was 50 ± 2 kg/m^3, the corresponding bending strength was 465.9 kPa and the 10 % compression strength was 479.5 kPa, which was related to the improvement of the bubble wall thickness in the TMfoam at a higher Ad.

Furthermore, the test curves of the mechanical properties of different TMfoam samples were compared to explain the law of foam toughness change, as shown in Fig. 8.

From Fig. 8, the bending strength and 10 % compression strength test curves of the TMfoam samples show the characteristics of phased changes, especially the compression strength curve changes most obviously. This indicates that foam samples with different dosage of HPA have different properties in response to external stress changes.

In Fig. 8a, with "★" as the turning point, the curve of the bending strength test can be roughly divided into two linear change stages: the front and the back. The bending strength and the applied bending stress both show a proportional growth relationship. Before the turning point, the rigidity of the foam mainly plays a role in resisting the external bending stress. After the turning point, it is the flexible cushioning effect

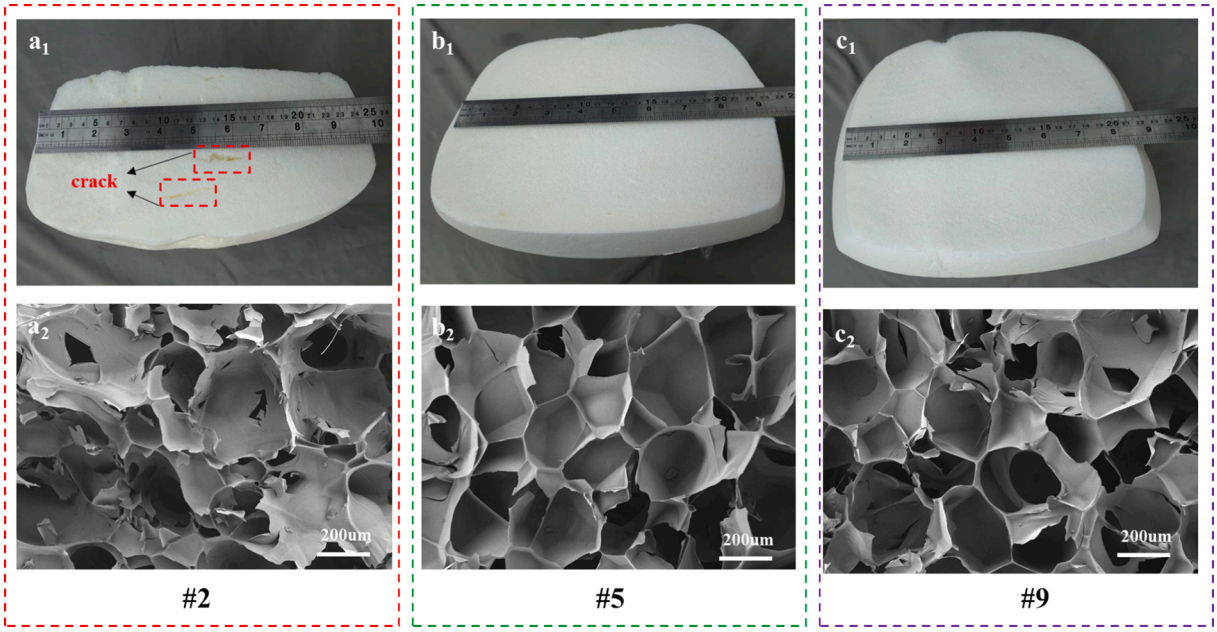

Fig. 6. Macroscopic and scanning electron micrographs of TMfoam.

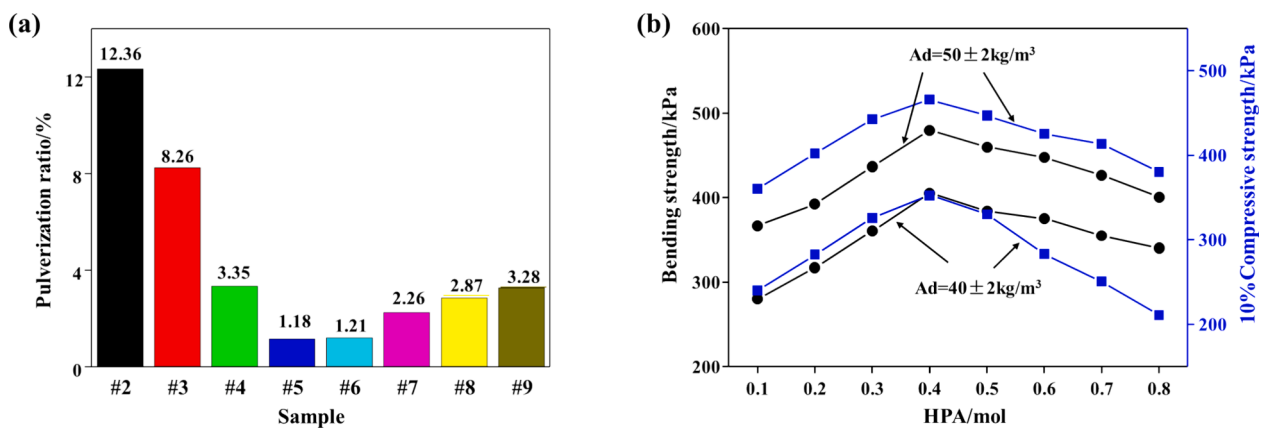

Fig. 7. The pulverization rate of the TMfoam (a); The basic mechanical properties of TMfoam (b).

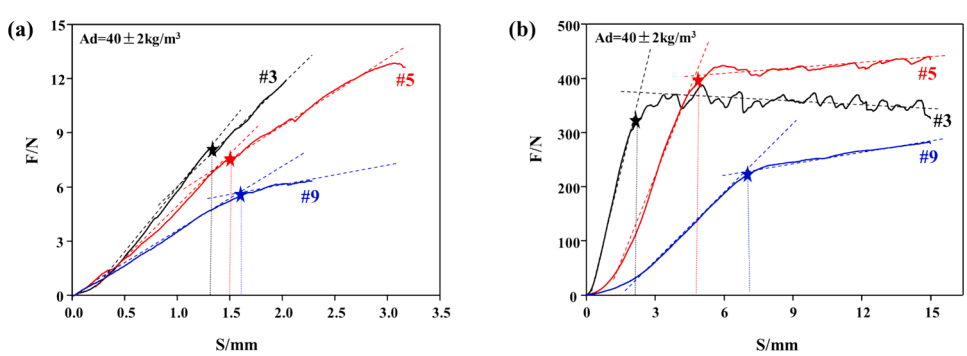

Fig. 8. Bending strength curve of TMfoam (a); Compression strength curves of the TMfoam (b).

of the foam, which effectively improves the rigidity of the foam itself when it is pressed too fast to break against the applied stress. The organic unity of flexibility and rigidity of the foams are beneficial to achieve the toughness of TMfoam necessary to resist the external bending stress.

In addition, after the turning point, compared with the TMfoam samples with lower HPA content, the test curve of bending strength for the TMfoam samples with higher HPA content decreased significantly, and the overall slope of the test curve gradually decreased. This once again shows that higher HPA dosage has a positive correlation with improving the flexibility of the TMfoam. It can be seen that the TMfoam (sample #5) has desired toughness performance, due to the organic

unity of rigidity and flexibility.

Similarly, in Fig. 8b, the compressive strength curve of the TMfoam can also be divided into rigid (before the turning point) and flexible (after the turning point) stages. Before the turning point, the rigidity of the foam mainly plays a role in resisting the external compressive stress. After the turning point, due to the flexible cushioning property, the simple rigidity is weakened correspondingly. When the foam was against the applied stress, it collapses and cracks too fast. At the same time, when the dosages of HPA are higher, the foam sample has relatively high flexibility, and the slope of the test curve will decrease significantly.

In summary, with the dosages of HPA increasing, the flexibility of TMfoam can be significantly improved. At the same time, a reasonable dosage of HPA can take into account and promote the organic unity of flexibility and rigidity of the TMfoam, and effectively improve the toughness, which is an important basis for the synergistic effect of DPR on the toughness of the TMfoam. Specifically, when the dosage of HPA is 0.4 mol (sample #5), the TMfoam possesses a relatively ideal overall physical structure, and the ratio of molecular weight and number of linear long chains in relatively reasonable condition. The number of intact bubble chamber is large, and the flexibility and rigidity of the foam unified perfectly, which gives the best toughness and strength of the TMfoam on the whole.

3.3.2. The overall performance of TMfoam

In addition to the pulverization rate and the basic mechanical properties of the TMfoam, the other important properties were also tested and compared, including thermal conductivity, flame retardancy, and sound absorption. The results are shown in Table 3.

As can be seen from Table 3, when the Ad of the TMfoam is 40 ± 2 kg/m^3, the thermal conductivity of the TMfoam with different dosages of HPA ranges from 0.031 to 0.033 W/(m·K), with horizontal combustion at the HB level, vertical combustion at the V-0 level, and a combustion grade of B. With the increase of HPA dosage, the LOI gradually decreases. When the dosages of HPA exceed to 0.6 mol, the combustion grade decreases from B_1 to B_2.

In terms of sound absorption performance, with the dosage increase of HPA, the peak value of the sound absorption coefficient first shifts to high frequencies and then to low frequencies, showing a trend of first increasing and then decreasing. The sound absorption performance of TMfoam (sample #5) is excellent, with the highest sound absorption coefficient reaching 0.92 at 3045 Hz, indicating that the TMfoam has the potential to be used as a sound absorption material [30–32].

From the above, it can be seen that when the dosage of HPA is 0.4 mol, the TMfoam prepared by the DPR have good structural toughness and other properties. When the Ad is 40 ± 2 kg/m^3, the bending strength can reach to 405.2 kPa, the 10 % compression strength is 352.5 kPa, the pulverization rate is 1.18 %, the combustion grade is B_1, the thermal conductivity is 0.03283 W/(m·K), and the sound absorption coefficient at 3045 Hz can reach 0.92, which has excellent overall performance.

3.3.3. The forming mechanism of the TMfoam

Based on the above experimental data and discussion, the mechanism about the synergistic effect of DPR on improving the structural toughness of TMfoam was analyzed.

Through the synergistic effect of the DPR, the foaming process of the TMfoam can be divided into two stages: the foam forming and the foam setting.

(1) Foam forming stage: Starting from the forming temperature (50 °C), the foam solution undergoes the formation and growth of TMfoam until the volume of the bubble chamber was maximized, achieving the initial stability process of the TMfoam. The progressive foam forming process is accompanied by the polycondensation reaction of hydroxymethyl groups in the Moli and the subsequent free radical polymerization reaction of vinyl groups. The two types of reactions start successively, being both independent and synergistic, jointly promoting the gradual increase in the molecular weight of the Moli to the maximum volume of the bubble chamber, forming a temperature-dependent metastable form of TMfoam. This stage is more complex and the key stage of regulation.
(2) Foam setting stage: At the setting temperature (130 °C), the TMfoam continues to polymerize, prompting the residual and reactive hydroxymethyl groups and vinyl groups in the first stage to fully participate in the polymerization reaction. The molecular weight of the Moli is maximized, the wall strength of the bubble is maximized, the bubble structure in the foam develops from metastable to stable, and the body and volume of the TMfoam is immobilized.

The above two stages are integrated into the preparation process of TMfoam, which are carried out successively and cannot be separated artificially. Through the rational control of polymerization conditions, the synergistic effect of the DPR can be realized, and TMfoam with good structural toughness and excellent properties can be obtained.

3.4. The main performance advantages of TMfoam

The TMfoam prepared in this paper was compared with other research and commercial application foams, such as polyurethane foam (PU-foam) and polypropylene foam (PP-foam), as shown in Table 4.

As can be seen from Table.4, the commercial PP-foam and PU-foam are combustible material, which has an apparent density (Ad) of 30–40 kg/m^3, and the LOI of (<21 %), the 10 % compressive strength and bending strength are lower than the TMfoam prepared in this paper.

MF-foam samples of literature [18,20,33] are B_1-Grade flame retardant materials with a LOI value in the range of 38 % to 40 %. Among these samples, the MF-foam1 in literature [18] has an Ad of 63 kg/m^3, which is 1.5 times that of the TMfoam of this work (Ad = 40 kg/m^3), along with the foam pulverization rate is 4.4 times that of the TMfoam. The MF-foam2 in literature [20] has an Ad of 250 kg/m^3, and its

Table 3
The overall performance of TMfoam.

Sample	Ad (kg/m$^{3)}$	Thermal conductivity W/(m.K)	LOI (%)	Combustion performance			Absorption performance	
				Horizontal	Vertical	Grade	Frequency (HZ)	Coefficient
#1	/	/	/	/	/	/	/	/
#2	39.6	0.03216	42.86	HB	V-0	B_1	1065	0.65
#3	40.8	0.03168	40.20	HB	V-0	B_1	1450	0.72
#4	41.3	0.03245	38.50	HB	V-0	B_1	2640	0.84
#5*	40.2	0.03283	35.40	HB	V-0	B_1	3045	0.92
#6	39.6	0.03239	31.19	HB	V-0	B_1	3250	0.90
#7	40.5	0.03281	29.68	HB	V-0	B_2	3545	0.88
#8	39.7	0.03244	27.54	HB	V-0	B_2	3320	0.82
#9	40.6	0.03186	26.51	HB	V-0	B_2	2320/3064	0.75/0.60

Note: Sample #1 cannot be foamed; * represents the optimum sample.

Table 4
The main performance advantages of TMfoam.

Foam type	Ad (kg/m$^{3)}$	LOI (%)	10 % compression strength (kPa)	Bending strength (kPa)	Thermal conductivity {W/(m·K)}	Pulverization rate (%)
PP-foam*	30–40	< 21	171.3	281.6	/	/
PU-foam*	30–40	< 21	148.8	223.1	/	/
MF-foam1 [18]	63	39.6	355.3	440	0.031	5.21
MF-foam2 [20]	250	38.9	5080	/	0.026	/
MF-foam3 [33]*	40	39.4	152	230	0.040	8.66
TMfoam (This work)	40 ± 2	38.5	352.5	405.2	0.033	1.18

Note: * represents the result of commercial foams measured under the same conditions.

practical value is extremely limited. The Ad and LOI of MF-foam3 in literature [33] are close to that of the TMfoam, but its mechanical properties are much lower, together with a foam pulverization rate is more than 7 times that of the TMfoam.

In a word, the TMfoam prepared in this study exhibits a more excellent performance than that of the thermo-plastic commercial foam, and the other reported MF-foams, and has a huge technical development value and commercial prospects.

4. Conclusion

In this paper, a novel melamine resin oligormer (Moli) with good foamability has been synthesized by the n(melamine):n (polyformaldehyde):n (hydroxypropyl acrylate) = 1:3.0:0.4, which contains vinyl and hydroxymethyl active groups. Furthermore, through the reasonable control of each component, a new melamine resin foam (toughned melamine resin foam, TMfoam) with excellent structural toughness and mechanical performance has been prepared by double polymerization reaction (DPR) during both the foam forming (50 °C) and foam setting (130 °C) stages. When the apparent density of the TMfoam is 40 ± 2 kg/m^3, the bending strength can reach to 405.2 kPa, the compression strength can reach to 352.5 kPa, and the pulverization rate is 1.18 %. The horizontal combustion is HB, the combustion grade reaches to B_1, and the thermal conductivity is 0.03283 W/(m·K). when the frequency is 3045 Hz, the absorption coefficient reaches to 0.92. The main properties of the toughned melamine resin foam prepared by DPR exceed those of commercial foams, showing good technical advantages and application prospects, and having great potential for commercial development.

Moreover, a mechanism of DPR was suggested. The preparation of TMfoam includes two stages: foam forming and the foam setting. First, the foam forming stage is a gradual process, accompanied by the polycondensation of hydroxymethyl groups in the Moli, and the subsequent free radical polymerization of vinyl groups, forming a temperature-dependent metastable TMfoam. Second, the foam setting stage is a process in which the remaining reactive active groups participate in the polymerization reaction, maximizing the molecular weight, maximizing the bubble wall strength, and fully curing and setting the entire TMfoam to achieve a steady state.

CRediT authorship contribution statement

Lijiu Cao: Writing – original draft, Project administration. **Yue Hu:** Writing – original draft, Methodology, Investigation. **Xu Ma:** Methodology, Investigation, Data curation. **Yunpeng Gong:** Methodology, Investigation, Data curation. **Xiuxiu Zhang:** Software, Resources, Investigation. **Jian Huang:** Resources, Investigation. **An Yan:** Methodology, Investigation. **Yufang Chen:** Writing – review & editing, Resources, Project administration. **Tao Jin:** Project administration, Funding acquisition.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Yufang Chen reports financial support was provided by Xinjiang Xue feng Sci-Tech (Group) Co., Ltd. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

References

[1] Y. Wang, Z. Chen, Y. Lu, et al., A review of application, modification, and prospect of melamine foam, Nanotechnol. Rev. 12 (2023) 20230137, https://doi.org/10.1515/ntrev-2023-0137.
[2] V. Nemanič, B. Zajec, M. Zumer, et al., Synthesis and characterization of melamine-formaldehyde rigid foams for vacuum thermal insulation, Appl. Energy 114 (2014) 320–326, https://doi.org/10.1016/j.apenergy.2013.09.071.
[3] X. Wang, Z. Han, Y. Liu, et al., Micro-nano surface structure construction and hydrophobic modification to prepare efficient oil-water separation melamine formaldehyde foam, Appl. Surf. Sci. (2019), https://doi.org/10.1016/j.apsusc.2019.144577.
[4] N. Amaly, A.Y. EL-Moghazy, N. Nitin, et al., Design, preparation, and application of novel multilayer metal-polyphenol composite on macroporous framework melamine foam for effective filtration removal of tetracycline in fluidic systems, Sep. Purif. Technol. 321 (2023) 124238, https://doi.org/10.1016/J.SEPPUR.2023.124238.
[5] L. Cao, J. Zhang, X. Zhang, et al., Stearyl chloride modified melamine derivatives and their resinification applications, Fine Chem. 04 (40) (2023) 812–819, https://doi.org/10.13550/j.jxhg.20220861.
[6] C. Li, H. Ma, C. Song, et al., Preparation and performance of melamine-formaldehyde rigid foams with high closed cell content, Cell. Polym. (2021), https://doi.org/10.1177/02624893211017130.
[7] M. Kavšek', N. Figar', I. Mihelič, et al., Melamine-formaldehyde rigid foams –Manufacturing and their thermal insulation properties, J. Cell. Plast. (2021) 1–19, https://doi.org/10.1177/0021955X21997348.
[8] H. Zhu, S. Xu, Preparation of flame-retardant rigid polyurethane foams by combining modified melamine−formaldehyde resin and phosphorus flame retardants, ACS Omega 5 (2020) 9658–9667, https://doi.org/10.1021/acsomega.9b03659.
[9] F. Luo, Wu. Kun, H. Zhu, et al., Intrinsically flame retarded foams based on melamine-formaldehyde condensates: thermal and mechanical properties, Polym. Int. (2017), https://doi.org/10.1002/pi.5314.
[10] N.N. Sakhadeo, T. Umasankar Patro, Exploring the multifunctional applications of surface-coated polymeric foams-a review, Ind. Eng. Chem. Res. 61 (2022) 5366–5387, https://doi.org/10.1021/acs.iecr.1c04945.
[11] H. Zhu, S.-A. Xua, Synthesis and properties of rigid polyurethane foams synthesized from modified urea-formaldehyde resin, Constr. Build. Mater. 202 (2019) 718–726, https://doi.org/10.1016/j.conbuildmat.2019.01.035.
[12] D. Xu, K. Yu, K. Qian, Thermal degradation study of rigid polyurethane foams containing tris(1chloro-2-propyl)phosphate and modified aramid fiber, Polym. Test. 67 (159–168) (2018), https://doi.org/10.1016/j.polymertesting.2018.01.034.
[13] Y. Shen, J. Gu, H. Tan, et al., Preparation and properties of a polyvinyl alcohol toughened urea-formaldehyde foam for thermal insulation applications, Constr. Build. Mater. 120 (2016) 104–111, https://doi.org/10.1016/j.conbuildmat.2016.05.096.
[14] T. Ge, K. Tang, X. Tang, Preparation and properties of acetoacetic ester-terminated polyether pre-synthesis modified phenolic foam, Materials (Basel) 12 (3) (2019) 334, https://doi.org/10.3390/ma12030334.
[15] S. Zhang, W. Ji, Y. Han, Flame-retardant expandable polystyrene foams coated with ethanediol-modified melamine-formaldehyde resin and microencapsulated ammonium polyphosphate, J. Appl. Polym. Sci. 135 (2018) 46471, https://doi.org/10.1002/app.46471.
[16] X. Xia, X. Gu, L. Feng, et al., Advances on toughening modification of melamine foam, Polym. Mater. Sci. Eng. 35 (04) (2019), https://doi.org/10.16865/j.cnki.1000-7555.2019.0121.

[17] Y. Feng, Qu. Wei, M. Liu, et al., Research review on toughening modification of melamine formaldehyde resin, Chin. J. Wood Sci. Technol. 36 (01) (2022), https://doi.org/10.12326/j.2096-9694.2021043.
[18] C. Li, H. Ma, Z. Zhou, et al., Preparation and properties of melamine-formaldehyde rigid closed-cell foam toughened by ethylene glycol/carbon fiber, Cell. Polym. (2020) 1–18, https://doi.org/10.1177/0262489320929232.
[19] W.-H. Li, J. Zhang, C. Yang, et al., Preparation and spectral analysis of melamine-formaldehyde resin modified by benzoguanamine, Spectrosc. Spectr. Anal. 38 (05) (2018), https://doi.org/10.3964/j.issn.1000-0593(2018)05-1481-05.
[20] C. Yan, Y.-J. Luo, W.-G. Zhang, Preparation of a novel melamine foam structure and properties, J. Appl. Polym. Sci. 139 (2020) 1–10, https://doi.org/10.1002/app.51992.
[21] Hu. Jiangbing, J. Sun, K. Qian, et al., Cellulose nanocrystals reinforced rigid melamine formaldehyde foam: preparation and property, J. Mater. Sci. Eng. 37 (02) (2019), https://doi.org/10.14136/j.cnki.issn1673-2812.2019.02.019.
[22] J.-L. Wang, J.-Q. Hou, Xu. Heng, et al., Preparation and performance characterization of tripolycyanamide formaldehyde foam board modified by PTMEG, J. Funct. Polym. 31 (04) (2018), https://doi.org/10.14133/j.cnki.1008-9357.20171226001.
[23] D. Wang, X. Zhang, D. Yang, et al., Study of preparation and properties of fire-retardant melamine formaldehyde resin foam, Adv. Mat. Res. 510 (2012) 634–638, https://doi.org/10.4028/www.scientific.net/AMR.510.634.
[24] C. Li, H. Ma, Z. Zhou, et al., Preparation and properties of melamine-formaldehyde rigid closed-cell foam toughened by ethylene glycol/carbon fiber, Cell. Polym. (2020), https://doi.org/10.1177/0262489320929232.
[25] S. Weiss, K. Urdl, H.A. Mayer, et al., IR spectroscopy: Suitable method for determination of curing degree and crosslinking type in melamine-formaldehyde resins, J. Appl. Polym. Sci. (2019), https://doi.org/10.1002/app.47691.
[26] R. Seidl, S. Weiss, R.W. Kessler, et al., Prediction of residual curing capacity of melamine-formaldehyde resins at an early stage of synthesis by in-line FTIR spectroscopy, Polymers 13 (2021) 2541, https://doi.org/10.3390/polym13152541.
[27] X. Wang, L. Cao, Y. Hu, et al., Highly transparent, hydrophobic, hard and flexible coatings based on a novel melamine-formaldehyde resin synthesized by hydrophobic melamine, Prog. Org. Coat. 179 (2023) 07487, https://doi.org/10.1016/j.porgcoat.2023.107487.
[28] G. Zhang, Y. Liu, J. Zhao, et al., Radical cascade reactions of unsaturated C-C bonds involving migration, Sci. China Chem. 62 (2019), https://doi.org/10.1007/s11426-019-9629-x.
[29] C. Chatgilialoglu, S. Barata-Vallejo, T. Gimisis, et al., Radical reactions in organic synthesis: exploring in-, on-, and with-water methods, Molecules 29 (2024) 569, https://doi.org/10.3390/molecules29030569.
[30] Y. Fei, W. Fang, M. Zhong, et al., Extrusion foaming of lightweight polystyrene composite foams with controllable cellular structure for sound absorption application, Polymers 11 (2019) 106, https://doi.org/10.3390/polym11010106.
[31] Y. Liu, X. Zhao, L. Ye, A novel elastic urea-melamine-formaldehyde foam: structure and properties, Proper. Ind. Eng. Chem. Res. (2016), https://doi.org/10.1021/acs.iecr.6b01957.
[32] S. Jiang, J. Zeng, H. Zhu, et al., Nanocellulose/polyvinylidene fluoride foam for high efficiency acoustic absorption at low and medium frequencies, J. Cell. Sci. Technol. (2023), https://doi.org/10.16561/j.cnki.xws.2023.04.01.
[33] Y. Tang, S. Wang, G. Li, Property and application of melamine rigid foam, New Chem. Mater. 48 (2020) 146–149, https://doi.org/10.19817/j.cnki.issn1006-3536.2020.S.034.